AIMS DTU Research Intern 2026

Audio-Visual Approximation of Video Semantic Space

Preview

The primary objective of this task is to design, train, and evaluate a multimodal representation learning framework. The candidate must determine whether the semantic embedding of a complete video sequence can be accurately approximated using only partial multimodal inputs, specifically, a single static visual frame (or sparse keyframes) and its corresponding audio track. The goal is to achieve competitive retrieval performance while significantly reducing the computational overhead associated with dense temporal video processing.

Your Assignment

The candidate is expected to develop an end-to-end pipeline encompassing the following three stages:

1. Feature Extraction & Mapping: Process an established multimodal dataset to extract independent audio and visual (image) embeddings. Map these disparate modalities into a shared semantic latent space designed to approximate ground-truth full-video embeddings.

2. Architectural Comparison: Implement and evaluate at least two distinct representation learning methodologies for this latent approximation. Acceptable strategies include, but are not limited to:
   - Contrastive alignment (e.g., InfoNCE loss across modalities)
   - Latent space regression (predicting the target video vector)
   - Cross-modal distillation (using a pre-trained video model as a teacher)
   - Transformer-based multimodal fusion

3. Evaluation & Benchmarking: Test the synthesized embeddings on standard downstream semantic tasks (such as cross-modal retrieval, similarity matching, or zero-shot classification) to quantify the viability of the approximation.

Evaluation

The proposed models must be evaluated across three primary dimensions to assess accuracy, practical utility, and computational efficiency:

A. Latent Space Proximity - How accurately does the synthesized embedding map to the target video embedding?

- Cosine Similarity: The primary metric for measuring the angular distance between the predicted audio-visual embedding and the ground-truth video embedding
- Mean Squared Error (MSE): Utilized if direct vector regression is implemented

B. Downstream Task Efficacy (Retrieval) - Is the approximated embedding practically viable for semantic search?

- Recall@K (R@1, R@5, R@10): The percentage of queries where the correct target video is retrieved within the top K results
- Median Rank (MedR): The median position of the target video across all retrieval queries

C. Computational Efficiency - What is the tangible benefit of bypassing temporal processing?

- Inference Latency: Measured in milliseconds per sample, comparing the audio-visual approximation method against a standard full-video processing baseline
- Parameter Count & FLOPs: A quantification of the model's computational footprint

Recommended Baselines for Comparison:

Baseline 1: Naive Late Fusion (The Architectural Baseline) - Extract features using frozen, off-the-shelf unimodal encoders (e.g., standard CLIP for the image frame and AST/VGGish for the audio). Concatenate the two vectors and pass them through a simple Multi-Layer Perceptron (MLP) to project them into the target video embedding dimension. Purpose: This serves as the absolute minimum viable product. Any advanced fusion strategy must demonstrably outperform this simple concatenation.

Baseline 2: ImageBind (Meta AI) or AudioCLIP (The State-of-the-Art Benchmark) - ImageBind is a highly relevant existing model that learns a single joint embedding space spanning images, text, audio, and video. Purpose: Because ImageBind naturally aligns image+audio into the same space as video without requiring explicit temporal processing during retrieval, it represents the upper bound/state-of-the-art for this specific problem. The candidate is not expected to train a model of this scale, but they should utilize its zero-shot capabilities as a benchmark to see how close a lightweight, custom-trained approximation can get to a massive foundation model.

Deliverables

1. Complete technical report covering task objective, methodology requirements, evaluation metrics, results, interpretation, and comparative analysis against the recommended baselines.

2. GitHub repository with clean, reproducible code for the complete pipeline including feature extraction, model training, and evaluation frameworks.

3. All trained models or links to model weights stored in Google Drive with corresponding documentation.

4. Detailed comparison results showing performance on all three evaluation dimensions against both baselines.